Chapter 5. Seasonal Time Series Models

Seasonal components and seasonal ARIMA

Contents

5.1 Time series with a seasonal component **2**

5.2 Time series with trend and seasonality **2**

5.3 Seasonal ARIMA models **3**

5.4 Multiplicative seasonal models **4**

5.5 Selected exercises **6**

Many series repeat a similar pattern with a known period s: monthly data typically have $s = 12$, quarterly data $s = 4$. Such periodic behaviour is called *seasonality* (a seasonal effect), and the period s is assumed known from the nature of the data. This chapter adapts the ARIMA models of Chapter 4 to seasonal data and introduces the multiplicative seasonal model of Box and Jenkins.

5.1 Time series with a seasonal component

Motivation. A deterministic-looking periodic component makes a series nonstationary in a way that ordinary differencing does not remove; differencing at the seasonal lag s does.

Suppose

$$X_t = S_t + Z_t, \tag{5.1}$$

where $\{S_t\}$ is a *seasonal component* of period s and $\{Z_t\}$ is stationary. Precisely, $S_t = S_{t+s}$ for all t; $S_1, \dots, S_s$ are independent with $ES_t = 0$ and $\mathrm{Var}(S_t) = \sigma_t^2 < \infty$; $EZ_t = 0$ with autocovariance $\gamma_Z(h)$; and $\{S_t\}$ and $\{Z_t\}$ are independent. Then

$$\begin{aligned} EX_{t+h}X_t &= E(S_{t+h} + Z_{t+h})(S_t + Z_t) = ES_{t+h}S_t + \gamma_Z(h) \\ &= \begin{cases} \sigma_t^2 + \gamma_Z(h), & h = 0, \pm s, \pm 2s, \dots, \\ \gamma_Z(h), & \text{otherwise.} \end{cases} \end{aligned}$$

The covariance depends on t through σ_t^2, so $\{X_t\}$ is nonstationary; and even when all σ_t^2 are equal the autocovariance does not decay along the lags $s, 2s, \dots$, so no causal ARMA model can describe $\{X_t\}$.

To remove the seasonal component introduce the *seasonal difference operator*

$$\nabla_s = 1 - B^s.$$

Applying it to (5.1) (*seasonal differencing*) gives

$$Y_t = (1 - B^s)X_t = X_t - X_{t-s} = S_t + Z_t - (S_{t-s} + Z_{t-s}) = Z_t - Z_{t-s},$$

a difference of a stationary series, hence stationary. Fitting an ARMA model $\phi(B)Y_t = \theta(B)\epsilon_t$, $\{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2)$, to Y_t we obtain

$$\phi(B)\nabla_s X_t = \theta(B)\epsilon_t. \tag{5.2}$$

Definition 5.1.1 (Seasonal difference model)

The model (5.2) is called the *seasonal difference model*.

5.2 Time series with trend and seasonality

Motivation. When a trend and a seasonal component appear together, one seasonal difference removes the seasonality and lowers the degree of the polynomial trend by one; ordinary differences then remove what is left of the trend.

Suppose

$$X_t = T_t + S_t + Z_t,$$

where T_t is a trend, S_t a seasonal component and Z_t a stationary process, mutually independent, with

$$T_t = \beta_0 + \beta_1 t + \cdots + \beta_r t^r \quad (r \geq 1,\ \beta_r \neq 0), \qquad S_t = S_{t+s} \text{ for all } t.$$

Seasonal differencing gives $\nabla_s X_t = \nabla_s T_t + \nabla_s S_t + \nabla_s Z_t$ with $\nabla_s S_t = 0$.

Lemma (Removing a polynomial trend)

With T_t and Z_t as above,
(1) $(1-B)^{r-1}\nabla_s T_t = \mu$ for the constant $\mu = s\,r!\,\beta_r$;
(2) $(1-B)^{r-1}\nabla_s Z_t$ is stationary with mean 0.
Consequently

$$(1-B)^{r-1}\nabla_s X_t = (1-B)^{r-1}\nabla_s T_t + (1-B)^{r-1}\nabla_s Z_t = \mu + (1-B)^{r-1}\nabla_s Z_t. \tag{5.3}$$

Proof. [Supplement]

Step 1. If $P(t) = at^k + (\text{lower terms})$ is a polynomial of degree $k \geq 1$, then for any integer $m \geq 1$, by the binomial theorem $t^k - (t-m)^k = km\,t^{k-1} + (\text{lower terms})$, so $P(t) - P(t-m)$ is a polynomial of degree $k-1$ with leading coefficient akm. (For $k = 0$ the difference is 0.)

Step 2. Apply Step 1 with $m = s$: $\nabla_s T_t$ has degree $r-1$ and leading coefficient $\beta_r rs$. Apply it $r-1$ more times with $m = 1$: the degree drops to 0 and the leading coefficient becomes $\beta_r rs \cdot (r-1)(r-2)\cdots 1 = s\,r!\,\beta_r$. This proves (1).

Step 3. $(1-B)^{r-1}\nabla_s = \sum_{j=0}^{m} c_j B^j$ is a polynomial in B of finite degree $m = r-1+s$. Hence $U_t = \sum_j c_j Z_{t-j}$ has $EU_t = 0$ and $\mathrm{Cov}(U_{t+h}, U_t) = \sum_{j,k} c_j c_k \gamma_Z(h-j+k)$, which does not depend on t. This proves (2), and (5.3) follows by linearity and $\nabla_s S_t = 0$. ■

Fitting an ARMA model to the stationary series $(1-B)^{r-1}\nabla_s X_t - \mu$ gives

$$\phi(B)\{(1-B)^{r-1}\nabla_s X_t - \mu\} = \theta(B)\epsilon_t, \qquad \{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2). \tag{5.4}$$

Definition 5.2.1 (Seasonal difference model with trend)

The model (5.4) is called the *seasonal difference model with trend*.

For a linear trend ($r = 1$) a single seasonal difference suffices; in practice one often applies ∇_s and then one ordinary difference ∇ to reach stationarity.

5.3 Seasonal ARIMA models

Motivation. Instead of a fixed periodic component, seasonality can be stochastic: an observation depends on the observation one season earlier. ARMA recursions in the seasonal lag B^s model this.

The model

$$X_t = \phi X_{t-s} + \epsilon_t, \qquad \{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2), \tag{5.5}$$

takes seasonality into account. If $|\phi| < 1$, iterating gives the stationary solution

$$X_t = \sum_{i=0}^{\infty} \phi^i \epsilon_{t-si}. \tag{5.6}$$

This $\{X_t\}$ has no seasonal component $\{S_t\}$, yet it shows a seasonal effect of period s: $\gamma_X(h) = 0$ unless h is a multiple of s, so only observations a whole number of seasons apart are correlated. Such data are described by the *seasonal ARMA* (SARMA) model. Since the series may also have a seasonal trend, we allow seasonal differencing and obtain the seasonal ARIMA (SARIMA) model.

Definition 5.3.1 (SARIMA$(P,D,Q)_s$ model)

A time series $\{X_t\}$ follows a *seasonal ARIMA model* SARIMA$(P,D,Q)_s$ if

$$\Phi(B^s)\{(1-B^s)^D X_t - \mu\} = \Theta(B^s)\epsilon_t, \qquad \{\epsilon_t\} \sim \text{IID}(0,\sigma^2), \tag{5.7}$$

where

$$\Phi(z) = 1 - \Phi_1 z - \cdots - \Phi_P z^P, \qquad \Theta(z) = 1 + \Theta_1 z + \cdots + \Theta_Q z^Q$$

satisfy the causality and invertibility conditions. Usually $\mu = 0$ is assumed.

The model of Definition 5.3.1 is appropriate when only observations in the same position of the season are correlated; $D \neq 0$ is appropriate when the seasonal part carries a trend. Note that (5.4) and (5.7) are different models, and which one is preferable depends on the data.

Example 5.3.2 (ACF of seasonal MA and AR models)

(1) For the MA$(2)_s$ model $X_t = \epsilon_t + \theta_1\epsilon_{t-s} + \theta_2\epsilon_{t-2s}$, the autocovariance is that of an MA(2) placed on the lags $0, s, 2s$:

$$\gamma_X(h) = \begin{cases} (1+\theta_1^2+\theta_2^2)\sigma^2, & h = 0, \\ (\theta_1 + \theta_1\theta_2)\sigma^2, & h = s, \\ \theta_2\sigma^2, & h = 2s, \\ 0, & \text{otherwise } (h \geq 0). \end{cases}$$

(2) For the AR$(1)_s$ model $X_t - \phi X_{t-s} = \epsilon_t$, $|\phi| < 1$, (5.6) gives $\gamma_X(sj) = \sigma^2 \sum_{i\geq 0} \phi^i \phi^{i+j} = \phi^j \sigma^2/(1-\phi^2)$ and $\gamma_X(h) = 0$ if $s \nmid h$. Hence

$$\rho(h) = \begin{cases} \phi^{h/s}, & h = 0, s, 2s, \ldots, \\ 0, & \text{otherwise.} \end{cases}$$

5.4 Multiplicative seasonal models

Motivation. Seasonal data are correlated not only across seasons but also between neighbouring observations. Multiplying a seasonal ARIMA operator in B^s by an ordinary ARIMA operator in B captures both kinds of dependence with few parameters.

Definition 5.4.1 (Multiplicative seasonal ARIMA model)

A time series $\{X_t\}$ follows a *multiplicative seasonal* ARIMA$(p,d,q)(P,D,Q)_s$ model if

$$\phi(B)\Phi(B^s)(1-B)^d(1-B^s)^D X_t = \theta(B)\Theta(B^s)\epsilon_t, \qquad \{\epsilon_t\} \sim \text{IID}(0,\sigma^2),$$

with ϕ, θ as in Definition 4.1.1 and Φ, Θ as in Definition 5.3.1.

This model contains those of Definitions 5.1.1, 5.2.1 and 5.3.1 as special cases. The ACF of the stationary part combines the two scales: a short-lag pattern near 0 governed by ϕ, θ, repeated around the seasonal lags $s, 2s, \ldots$ governed by Φ, Θ.

Example 5.4.2 (The ARIMA$(0,1,1)(0,1,1)_s$ model)

The model $\nabla_s(1-B)X_t = (1+\theta B)(1+\Theta B^s)\epsilon_t$ reads

$$X_t - X_{t-1} - X_{t-s} + X_{t-s-1} = \epsilon_t + \theta\epsilon_{t-1} + \Theta\epsilon_{t-s} + \theta\Theta\epsilon_{t-s-1}.$$

Let $W_t = \nabla_s(1-B)X_t$; then $\{W_t\}$ is the stationary MA process ARIMA$(0,0,1)(0,0,1)_s$. Assume $s \geq 3$. Multiplying out the four noise terms and keeping the matching lags,

$$\gamma_W(0) = (1+\theta^2)(1+\Theta^2)\sigma^2, \qquad \gamma_W(1) = \theta(1+\Theta^2)\sigma^2,$$
$$\gamma_W(s) = \Theta(1+\theta^2)\sigma^2, \qquad \gamma_W(s-1) = \gamma_W(s+1) = \theta\Theta\sigma^2,$$

and $\gamma_W(h) = 0$ for $2 \leq h \leq s-2$ and for $h \geq s+2$ (Figure 1).

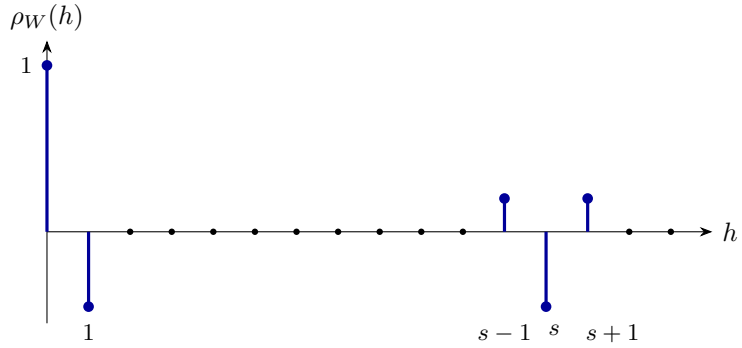


Figure 1: ACF of $W_t = (1+\theta B)(1+\Theta B^s)\epsilon_t$ (here $\theta, \Theta < 0$, $s = 12$): nonzero only at lags $0, 1, s-1, s, s+1$.

Example 5.4.3 (The ARIMA$(0,0,1)(1,0,1)_s$ model)

Consider $(1-\Phi B^s)X_t = (1+\theta B)(1+\Theta B^s)\epsilon_t$ with $|\Phi| < 1$ and $s \geq 4$. Write $X_t = (1+\theta B)Y_t$, where $Y_t = \Phi Y_{t-s} + \epsilon_t + \Theta\epsilon_{t-s}$ is a SARMA$(1,1)_s$ process. Since $\mathrm{Cov}(Y_t, \epsilon_t) = \sigma^2$ and $\mathrm{Cov}(Y_t, \epsilon_{t-s}) = (\Phi+\Theta)\sigma^2$,

$$\gamma_Y(0) = \Phi^2\gamma_Y(0) + (1+\Theta^2)\sigma^2 + 2\Phi\Theta\sigma^2 \implies \gamma_Y(0) = \Big\{1 + \frac{(\Theta+\Phi)^2}{1-\Phi^2}\Big\}\sigma^2,$$
$$\gamma_Y(s) = \Phi\gamma_Y(0) + \Theta\sigma^2 = \Big\{\Theta + \Phi + \frac{\Phi(\Theta+\Phi)^2}{1-\Phi^2}\Big\}\sigma^2,$$

$\gamma_Y(ks) = \Phi^{k-1}\gamma_Y(s)$ for $k \geq 1$, and $\gamma_Y(h) = 0$ if $s \nmid h$. The filter $1+\theta B$ gives $\gamma_X(h) = (1+\theta^2)\gamma_Y(h) + \theta\{\gamma_Y(h-1) + \gamma_Y(h+1)\}$, hence

$$\gamma_X(0) = (1+\theta^2)\Big\{1 + \frac{(\Theta+\Phi)^2}{1-\Phi^2}\Big\}\sigma^2, \qquad \gamma_X(1) = \theta\Big\{1 + \frac{(\Theta+\Phi)^2}{1-\Phi^2}\Big\}\sigma^2,$$
$$\gamma_X(s) = (1+\theta^2)\gamma_Y(s), \qquad \gamma_X(s-1) = \gamma_X(s+1) = \theta\gamma_Y(s),$$

$\gamma_X(2) = \cdots = \gamma_X(s-2) = 0$, and $\gamma_X(h) = \Phi\gamma_X(h-s)$ for $h \geq s+2$. Compared with Example 5.4.2, the seasonal AR factor makes the pattern around s recur, damped by Φ, around $2s, 3s, \ldots$.

5.5 Selected exercises

Motivation. The exercises below compute the causal representation and the autocovariances of seasonal models directly from the operator identities, and compare a seasonal MA model with an unrestricted MA model of the same order.

Exercise 5.2 (ψ-weights of a seasonal model)

Let $(1-\Phi B^s)X_t = (1+\theta B)\epsilon_t$ with $|\Phi|<1$ and $\{\epsilon_t\} \sim \mathrm{IID}(0,\sigma^2)$. Find the coefficients a_j in the representation $X_t = \sum_{j=0}^{\infty} a_j \epsilon_{t-j}$ of the stationary solution.

Solution.

Step 1. The AR polynomial $\phi(z) = 1-\Phi z^s$ vanishes only where $z^s = 1/\Phi$, i.e. on the circle $|z| = |\Phi|^{-1/s} > 1$ (no zeros at all if $\Phi = 0$). By Proposition 2.4.A, the model has a unique stationary solution $X_t = \sum_j a_j \epsilon_{t-j}$, whose coefficients are those of the power series $a(z) = (1+\theta z)/(1-\Phi z^s)$.

Step 2. For $|z| < |\Phi|^{-1/s}$, the geometric series gives $(1-\Phi z^s)^{-1} = \sum_{i\geq 0} \Phi^i z^{si}$, so

$$a(z) = \sum_{i=0}^{\infty} \Phi^i z^{si} + \theta \sum_{i=0}^{\infty} \Phi^i z^{si+1}.$$

Step 3: $s \geq 2$. The exponents si and $si'+1$ never coincide, since $s(i-i') = 1$ is impossible for $s \geq 2$. Comparing coefficients,

$$a_{si} = \Phi^i, \qquad a_{si+1} = \theta\Phi^i \quad (i = 0,1,2,\dots), \qquad a_j = 0 \text{ if } j \bmod s \notin \{0,1\}.$$

Thus $X_t = \sum_{i\geq 0} \Phi^i(\epsilon_{t-si} + \theta\epsilon_{t-si-1})$.

Step 4: $s = 1$. Then the model is an ordinary ARMA(1,1) and the two series overlap: $a_0 = 1$ and $a_j = \Phi^j + \theta\Phi^{j-1} = \Phi^{j-1}(\Phi+\theta)$ for $j \geq 1$.

Step 5: check. In both cases the coefficients satisfy $a_j - \Phi a_{j-s} = \mathbb{1}\{j=0\} + \theta\,\mathbb{1}\{j=1\}$ (with $a_j = 0$ for $j<0$), which is the coefficient form of $(1-\Phi z^s)a(z) = 1+\theta z$. ■

Exercise 5.3 ($\mathrm{MA}(1)_{12}$ versus $\mathrm{MA}(12)$)

Can the $\mathrm{MA}(1)_{12}$ model $X_t = \epsilon_t + \Theta\epsilon_{t-12}$ be regarded as an MA(12) model? Which of the two formulations is preferable, and why?

Solution.

Step 1: it is an MA(12) model. $X_t = \epsilon_t + \theta_1\epsilon_{t-1} + \cdots + \theta_{12}\epsilon_{t-12}$ with $\theta_1 = \cdots = \theta_{11} = 0$ and $\theta_{12} = \Theta$. Its autocovariances are those of Example 5.3.2(1) with one seasonal coefficient:

$$\gamma(0) = (1+\Theta^2)\sigma^2, \qquad \gamma(12) = \Theta\sigma^2, \qquad \gamma(h) = 0 \text{ for } h \geq 1,\ h \neq 12,$$

because $\mathrm{Cov}(\epsilon_{t+h} + \Theta\epsilon_{t+h-12}, \epsilon_t + \Theta\epsilon_{t-12})$ collects a product of equal-time noises only when $h \in \{0, \pm 12\}$.

Step 2: the same invertibility condition. The MA(12) polynomial $\theta(z) = 1+\Theta z^{12}$ vanishes exactly when $z^{12} = -1/\Theta$, i.e. on $|z| = |\Theta|^{-1/12}$. Hence all zeros lie outside the unit circle if and only if $|\Theta| < 1$, which is the invertibility condition of the seasonal $\mathrm{MA}(1)_{12}$ polynomial $1+\Theta z$. So the two descriptions define the same class of processes.

Step 3: which to prefer. As a model to be *fitted*, the unrestricted MA(12) has twelve free coefficients, eleven of which are zero for this process. Estimating them adds sampling error of order $n^{-1/2}$ to each

fitted coefficient, and hence to the fitted autocovariances and to the forecasts, without improving the fit: the likelihood ratio statistic for the eleven zero restrictions is asymptotically $\chi^2(11)$, whereas AIC charges $2 \cdot 11 = 22$ and BIC $11 \log n$ for the extra parameters. The seasonal MA$(1)_{12}$ model is therefore preferable by the principle of parsimony; it also states the structure directly (dependence only between the same month of consecutive years). The unrestricted MA(12) is preferable only if the sample ACF shows genuine correlation at lags $1, \dots, 11$, in which case a multiplicative model such as ARIMA$(0,0,1)(0,0,1)_{12}$ (Example 5.4.2) is usually tried first. ■

Exercise 5.4 (ACF of ARIMA$(1,0,1)(0,0,1)_s$)

The book asks for the autocovariance functions of four seasonal models; parts (1), (2) and (4) are worked out in Examples 5.4.3 (the process Y_t there), 5.4.2 and 5.4.3. Part (3): let $s \geq 2$, $|\phi| < 1$ and

$$(1 - \phi B)X_t = (1 + \theta B)(1 + \Theta B^s)\epsilon_t, \qquad \{\epsilon_t\} \sim \text{IID}(0, \sigma^2).$$

Find the autocovariance function of the stationary solution.

Solution.

Step 1: reduction to ARMA(1,1). Let Y_t be the causal stationary solution of $(1-\phi B)Y_t = (1+\theta B)\epsilon_t$ and put $X_t = Y_t + \Theta Y_{t-s}$. Since the filters commute,

$$(1 - \phi B)X_t = (1 - \phi B)Y_t + \Theta(1 - \phi B)Y_{t-s} = (1 + \theta B)\epsilon_t + \Theta(1 + \theta B)\epsilon_{t-s} = (1 + \theta B)(1 + \Theta B^s)\epsilon_t.$$

$\{X_t\}$ is stationary as a finite linear filter of a stationary series, and by Proposition 2.4.A (an ARMA$(1, s+1)$ with causal AR part) it is the unique stationary solution.

Step 2: ACVF of Y. By Example 2.4.3, with $c = \gamma_Y(1)$,

$$\gamma_Y(0) = \frac{\sigma^2(1 + 2\phi\theta + \theta^2)}{1 - \phi^2}, \qquad c = \frac{\sigma^2(\phi + \theta)(1 + \phi\theta)}{1 - \phi^2}, \qquad \gamma_Y(h) = c\,\phi^{|h|-1} \ (h \neq 0).$$

Step 3: general formula. By bilinearity of the covariance and stationarity of Y,

$$\begin{aligned} \gamma_X(h) &= \text{Cov}(Y_{t+h} + \Theta Y_{t+h-s},\, Y_t + \Theta Y_{t-s}) \\ &= (1 + \Theta^2)\gamma_Y(h) + \Theta\{\gamma_Y(h - s) + \gamma_Y(h + s)\}. \end{aligned}$$

Step 4: explicit values. Inserting Step 2 for $h \geq 0$ (and $\gamma_X(-h) = \gamma_X(h)$):

$$\begin{aligned} \gamma_X(0) &= (1 + \Theta^2)\gamma_Y(0) + 2\Theta c\,\phi^{s-1}, \\ \gamma_X(h) &= c\{(1 + \Theta^2)\phi^{h-1} + \Theta(\phi^{s-h-1} + \phi^{s+h-1})\}, \quad 1 \leq h \leq s - 1, \\ \gamma_X(s) &= (1 + \Theta^2)c\,\phi^{s-1} + \Theta\{\gamma_Y(0) + c\,\phi^{2s-1}\}, \\ \gamma_X(h) &= c\{(1 + \Theta^2)\phi^{h-1} + \Theta(\phi^{h-s-1} + \phi^{h+s-1})\}, \quad h \geq s + 1. \end{aligned}$$

In the last line every exponent increases by one when h does, so $\gamma_X(h) = \phi\,\gamma_X(h-1)$ for $h \geq s+2$: beyond the seasonal lag the ACF decays geometrically as for the AR(1) factor, while the term $\Theta\gamma_Y(0)$ produces the seasonal spike at $h = s$. The ACF is $\rho_X(h) = \gamma_X(h)/\gamma_X(0)$. ■